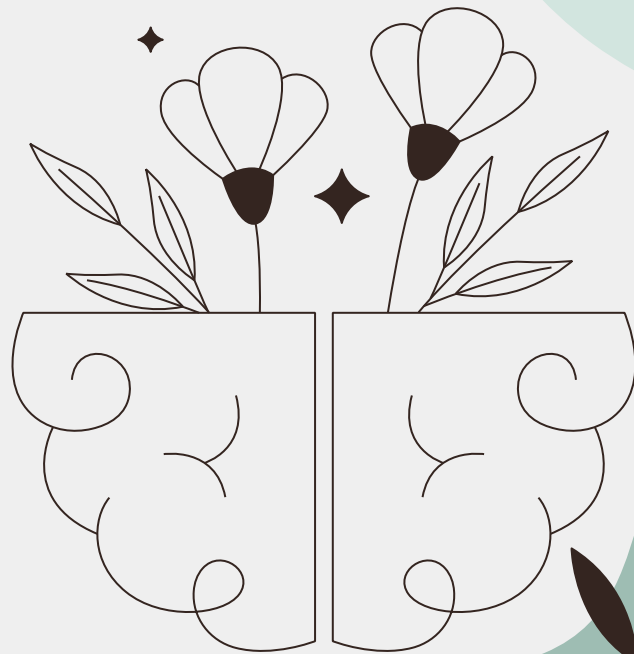




Mental Health Classifier

Using Mental Health Surveys to Predict Patient Condition





Manav Kumar

manavk05@berkeley.edu

Table of Contents

Background	Who, What, and Where of the dataset used
EDA/Meaningful Insights	Special Relationships, Difficulties, Set up Classifier
Forest Classifier	What is it & why I used it
Key Takeaways	Main conclusions from the results
My Growth	What I learned from this project

Project Goals

Background

Let's say you want to accurately tell if a person has a mental disorder. One central question to ask is: What factors can play into bad mental health? How can these factors predict mental health?

Dataset

The dataset chosen to use for this project is a survey taken by patients that classifies a person as Depression, Bipolar-Type 1, Bipolar-Type 2, or Normal

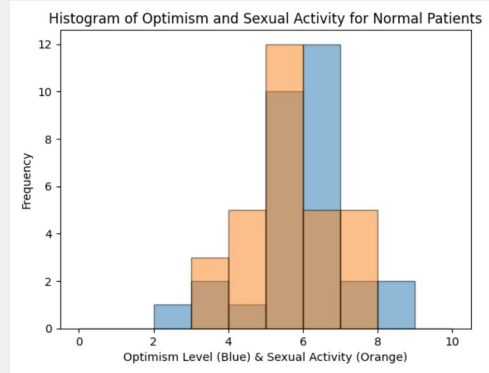
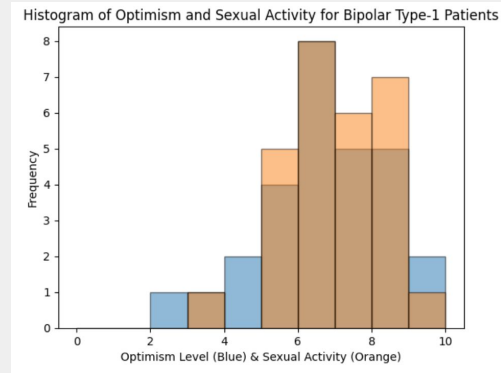
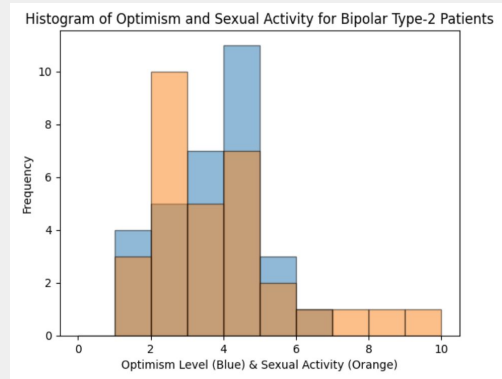
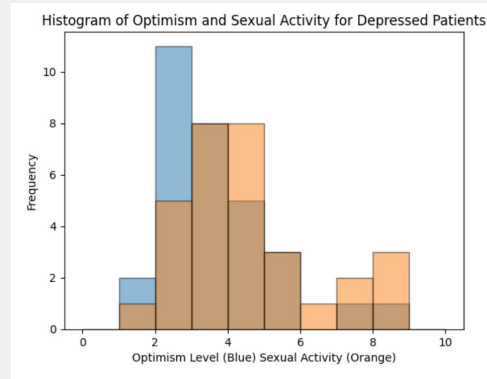
Significance

With mental health becoming more prevalent in today's society, I feel that an automated system would be beneficial for the healthcare industry by identifying early signs of mental health issues.

Exploratory Data Analysis

There was not too much EDA done here other than feature selection for the classifier. Some graphs were made to visualize the trends of certain variables.

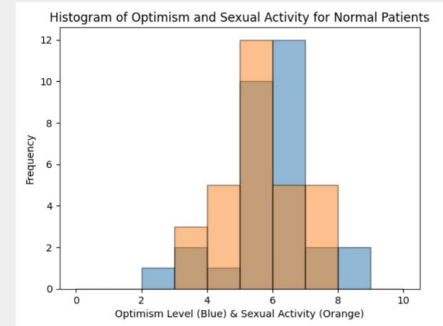
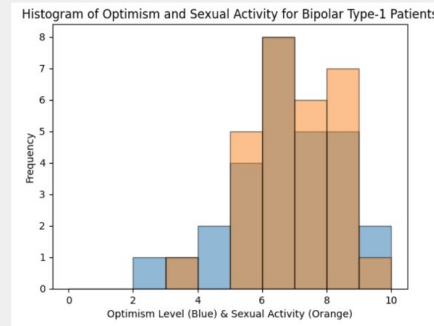
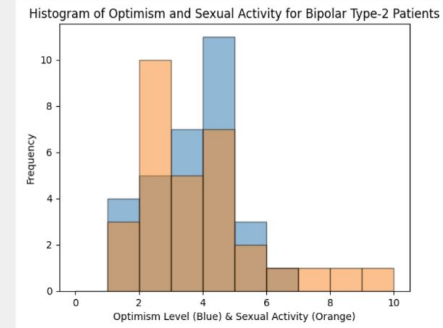
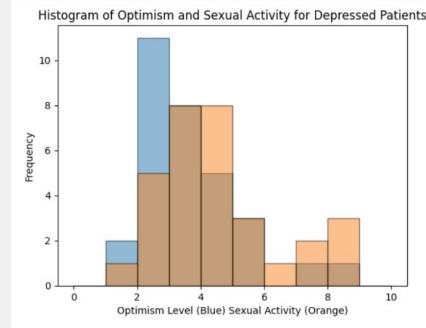
Each graph is for the type of patient (depressed, bipolar 1 or 2, and normal). Each graph has reported optimism level in Blue and reported frequency of Sexual Activity in Orange.



EDA Meaningful Insights

From the EDA, we are able to harness some meaningful trends in our data. In this case, we can see that optimism and sexual activity are somewhat linked within patients of the same class.

With the use of these trends, we are better able to interpret the results of the forest classifier and further analyze if necessary. Future analysis on this topic may benefit from a more in-depth EDA.



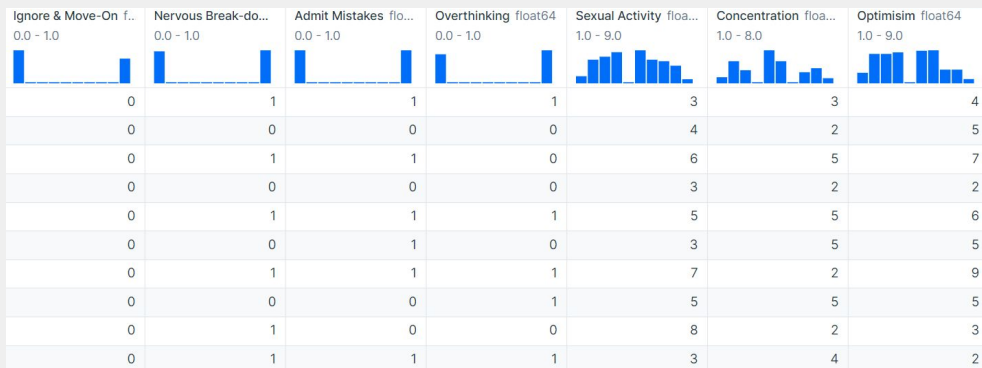
Meaningful Insights

Sadness object	Euphoric object	Exhausted object	Sleep disorder ob...
Usually 35%	Seldom 38.3%	Sometimes 31.7%	Sometimes 36.7%
Sometimes 35%	Sometimes 37.5%	Usually 28.3%	Usually 28.3%
2 others 30%	2 others 24.2%	2 others 40%	2 others 35%
Usually	Seldom	Sometimes	Sometimes
Usually	Seldom	Usually	Sometimes
Sometimes	Most-Often	Sometimes	Sometimes
Usually	Seldom	Usually	Most-Often
Usually	Usually	Sometimes	Sometimes
Usually	Sometimes	Sometimes	Most-Often
Seldom	Usually	Seldom	Sometimes
Usually	Sometimes	Sometimes	Sometimes
Most-Often	Seldom	Most-Often	Usually
Usually	Seldom	Most-Often	Sometimes

Although a one-hot-encoder would help with the process of changing these categorical variables into numeric variables, it makes the data much more difficult to work with, let alone make a forest classifier.

For these reasons I've chose to drop these four columns from the training data.

Feature Selection



- Features that were categorical have been discarded as the numerical and binary features were much smoother to work with in this dataset.
- Due to this decision, the final results of the forest regressor can be much different than it would have been if I had included the four categorical variables.
- Additionally, the binary values were converted from booleans.

Forest Classifier

Here a forest classifier was used since I wanted to be able to classify the mental disorder based on the variables I inputted, in other words use hypothetical data to classify one of four disorders.

I have worked with a forest regressor before as a team (see Airbnb project above this one on my website) and wanted to see if I could implement a classifier on my own this time, which is part of the reason I chose to work with this data.

The results of the classifier are slightly different than the regressor and these differences serve to display the use cases that each is unique to.

The main difference is that a classifier's final result is a categorical variable (in this case the type of disorder) while a regressor is numerical (price of Airbnb in the case of my other project).

Key Takeaways

I was able to successfully leverage the forest classifier with an accuracy of 75% in prediction.

```
Accuracy: 0.75
Classification Report:
              precision    recall  f1-score   support

Bipolar Type-1      0.50      0.50      0.50         4
Bipolar Type-2      0.83      1.00      0.91         5
    Depression      0.78      0.88      0.82         8
        Normal      0.80      0.57      0.67         7

   accuracy                   0.75        24
  macro avg      0.73      0.74      0.72        24
 weighted avg      0.75      0.75      0.74        24
```

Precision: Ratio of Correctly Predicted Positives to Total Predicted Positive

Recall Ratio: Ratio of Correctly Predicted Positives to All Actual Positive Instances

F1-Score: Harmonic Mean of Precision and Recall Ratio (Closer to 1 = Better)

Support: Number of times a certain class occurred

Accuracy: Ratio of total correct predictions to all predictions

Comments

- One of the main issues with this project is the very limited data, ideally thousands of rows would be ideal at the least.
- The decision to drop nonbinary and non numerical variables can also cause the classifier to behave differently compared to if all original data was kept.
- Overall, this was a rather fast project that I thought was interesting to see unfold.
- The essential thing I learned from this project was how to implement a forest classification model from scratch, something that I have had exposure to before, but have not done solo.
- In the future, I hope to implement this in the research paper I am working on at the moment with the Center for Targeted Machine Learning at UC Berkeley. They have some more complex ML approaches so I think my base knowledge of this project and some others will contribute to my understanding and utility on research.